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ChE 311 Midterm Exam I, Spring 2019

Closed book and closed notes. Only provided tables are allowed.

50 minutes, 100 points

1. (5 points) For the following velocity profile,

$$\vec{v} = (5t + 7x - 3y + z)\vec{\delta}_x + (x - 2y + z)\vec{\delta}_y + (5t + x - y + 5z)\vec{\delta}_z$$

The flow is incompressible.

$$7 - 2 + 5$$

(A) True

(B) False

(C) Not enough information to decide

2. (5 points) The flow defined by the following velocity profile with a cylindrical coordinate satisfies the continuity equation. For the flow, density is constant.

$$\vec{v} = (-5r)\vec{\delta}_r + (2t + r + 5z)\vec{\delta}_\theta + (5z)\vec{\delta}_z$$

(A) True

(B) False

(C) Not enough information to decide

$$\frac{1}{r} \frac{\partial}{\partial r} (r v_r) + \frac{1}{r} \frac{\partial}{\partial \theta} (r v_\theta) + \frac{\partial}{\partial z} (r v_z) = 0$$

$$\frac{1}{r} \frac{\partial}{\partial r} (-5r^2) + \frac{1}{r} \frac{\partial}{\partial \theta} (2r + r + 5z) + \frac{\partial}{\partial z} (5z) = 0$$

3. (10 points) Please choose the answer for the following questions.

$$\mathbf{u} = (3x_1 + 4x_2)\vec{\delta}_1 + (x_2 + 4x_3)\vec{\delta}_2 + (x_1 + 2x_3)\vec{\delta}_3$$

$$\frac{\partial v_1}{\partial x_1} + \frac{\partial v_2}{\partial x_2} + \frac{\partial v_3}{\partial x_3}$$

(1) $\nabla \cdot \mathbf{u} =$

(a) 16

(b) 6

(c) 10

(d) 5

(2) $\nabla \mathbf{u} =$

$$(a) \begin{vmatrix} 3 & 4 & 1 \\ 1 & 1 & 4 \\ 1 & 1 & 2 \end{vmatrix}$$

$$(b) \begin{vmatrix} 3 & 4 & 0 \\ 0 & 1 & 4 \\ 1 & 0 & 2 \end{vmatrix}$$

$$(c) \begin{vmatrix} 3 & 1 & 1 \\ 4 & 1 & 1 \\ 1 & 4 & 2 \end{vmatrix}$$

$$(d) \begin{vmatrix} 3 & 0 & 1 \\ 4 & 1 & 0 \\ 0 & 4 & 2 \end{vmatrix}$$

$$\begin{vmatrix} \frac{\partial v_1}{\partial x_1} & \frac{\partial v_2}{\partial x_1} & \frac{\partial v_3}{\partial x_1} \\ \frac{\partial v_1}{\partial x_2} & \frac{\partial v_2}{\partial x_2} & \frac{\partial v_3}{\partial x_2} \\ \frac{\partial v_1}{\partial x_3} & \frac{\partial v_2}{\partial x_3} & \frac{\partial v_3}{\partial x_3} \end{vmatrix}$$

4. (5 points) $\tau = \begin{vmatrix} 2x_1 + x_2 & x_2 + x_3 & 3x_1 + x_3 \\ x_1 & x_2 & x_3 \\ x_3 & x_2 & x_1 \end{vmatrix}$ $\frac{\partial}{\partial x_1} (2x_1 + x_2) + \frac{\partial}{\partial x_1} (x_1) x_3 + \frac{\partial}{\partial x_1} (x_3)$

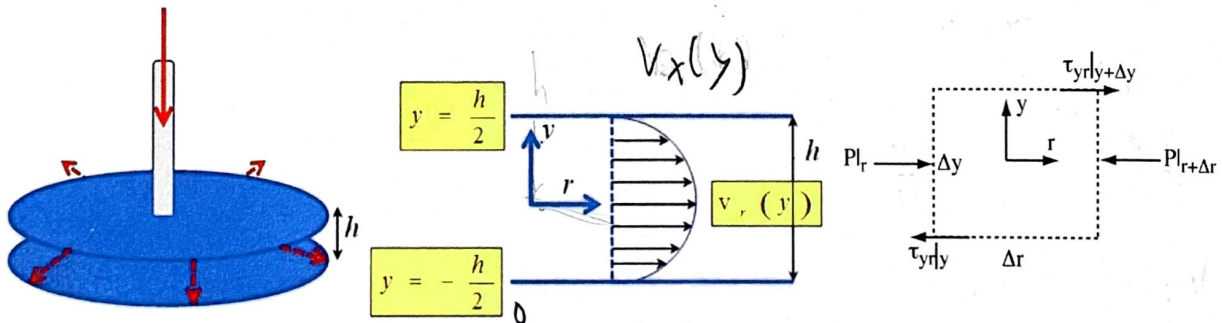
Please choose the right answer for $\nabla \cdot \tau$

(a) $\nabla \cdot \tau = \begin{vmatrix} 2 & 1 & 1 \\ 1 & 1 & 4 \\ 0 & 1 & 0 \end{vmatrix}$ (b) $\nabla \cdot \tau = \begin{vmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 4 & 0 \end{vmatrix}$

(c) $\nabla \cdot \tau = 3\delta_1 + \delta_2 + 3\delta_3$ (d) $\nabla \cdot \tau = 4\delta_1 + 3\delta_2 + \delta_3$

5. A radial-flow channel has been developed to measure the adhesion (energy) of cells to surfaces *in vitro*. The radial-flow channel is shown in the figure below. In this device, fluid flows in from a central source, flows out in a radial direction and the volumetric flow rate at any positions r is Q . The flow itself is driven by an unspecified pressure gradient.

- (1) Solve for the velocity profile (20 points).
- (2) Use your solution for the velocity profile and the constant flow rate condition to calculate the wall shear stress at $r = 1$ cm. The fluid viscosity is similar as water $1 \text{ mPa}\cdot\text{s}$. The volumetric flow rate is $10 \mu\text{L}/\text{min}$, with $h = 200 \mu\text{m}$. (20 points)



1) on the next page

2) $\tau_{y|y} = -\mu \frac{dv_r}{dy} = -\mu \left(\frac{P_r - P_{r+\Delta r}}{\mu \Delta r} \left(\frac{\partial}{\partial y} \left(\frac{y^2}{2} \right) \right) \right)$

$\mu = \frac{(P_r - P_{r+\Delta r})}{\Delta r} \left(y - \frac{h}{2} \right)$

at $t = 1 \text{ cm}$

$Q = \int_0^h \int_0^{\Delta r} v_z \, dr \, dy = \frac{P_r - P_{r+\Delta r}}{\mu \Delta r} \cdot \Delta r \cdot \int_0^h \left(\frac{hy}{2} - \frac{y^2}{2} + \frac{h^2}{4} \right) dy$

$= \frac{P_r - P_{r+\Delta r}}{\mu} \cdot \left(\frac{hy^2}{4} - \frac{y^3}{6} + \frac{h^2}{4}y \right) \Big|_0^h = \frac{P_r - P_{r+\Delta r}}{\mu} \cdot \left(\frac{h^3}{4} - \frac{h^3}{6} + \frac{h^3}{4} \right)$

$= \frac{P_r - P_{r+\Delta r}}{\mu} \cdot \left(\frac{3+3-2}{12} \right) h^3 = \frac{P_r - P_{r+\Delta r}}{\mu} \cdot \frac{1}{3} h^3 = \frac{10 \times 10^{-6} \text{ L}}{1 \text{ min}} \times \frac{1 \text{ min}}{60 \text{ s}} = 0.167 \times 10^{-6} \frac{\text{L}}{\text{s}}$

①



momentum flux:

$$\phi_{xx}|_x = (\tau_{yx}' + p + \rho v_x u_x) \cdot h \Delta x$$

$$\phi_{yx}|_{y=0} = (\tau_{yx}' + p' + \rho v_x v_y) \cdot L \Delta x$$

$$\phi_{zx}|_{z=0} = (\tau_{zx}' + p' + \rho v_z v_x) \cdot h \Delta x$$

should be r. θ. γ

you can not neglect curvature

momentum balance:

$$((p + \rho v_x u_x)|_x - (p + \rho v_x u_x)|_{x+\Delta x})) h L + (\tau_{yx}|_{y=0} - \tau_{yx}|_{y=h}) L \Delta x = 0$$

$$\frac{p_x - p_{x+\Delta x}}{\Delta x} = \frac{d\tau_{yx}}{dy}$$

$$\Rightarrow \tau_{yx} = \frac{p_x - p_{x+\Delta x}}{\Delta x} y + C_1$$

BC ①: at $y = \frac{h}{2}$, $\tau_{yx} = 0$ ✓

$$C_1 = -\frac{p_x - p_{x+\Delta x}}{\Delta x} \frac{h}{2}$$

$$\tau_{yx} = \frac{p_x - p_{x+\Delta x}}{\Delta x} (y - \frac{h}{2}) = -\mu \frac{\partial v_x}{\partial y}$$

$$\frac{\partial v_x}{\partial y} = \frac{p_x - p_{x+\Delta x}}{\mu \Delta x} (\frac{h}{2} - y)$$

$$v_x = \frac{p_x - p_{x+\Delta x}}{\mu \Delta x} (\frac{h}{2} y - \frac{y^2}{2}) + C_2$$

BC: at $y = -\frac{h}{2}$, $v_x = 0$ ✓

$$C_2 = \frac{p_x - p_{x+\Delta x}}{\mu \Delta x} (\frac{h^2}{4})$$

$$\Rightarrow v_x = \frac{p_x - p_{x+\Delta x}}{\mu \Delta x} (\frac{h}{2} y - \frac{y^2}{2} + \frac{h^2}{4}) \Rightarrow v_r = \frac{p_r - p_{r+\Delta r}}{\mu \Delta r} (\frac{h}{2} y - \frac{y^2}{2} + \frac{h^2}{4})$$

$$2) p_r - p_{r+\Delta r} = Q \cdot \frac{3}{4} \cdot \mu = 0.167 \times 10^{-6} \text{ N/m}^2 \times \frac{3}{(200 \times 10^{-6} \text{ m})^2} \times 1 \times 10^{-3} \text{ Pa} \cdot \text{s} \times \frac{10^{-3} \text{ m}^2}{14}$$

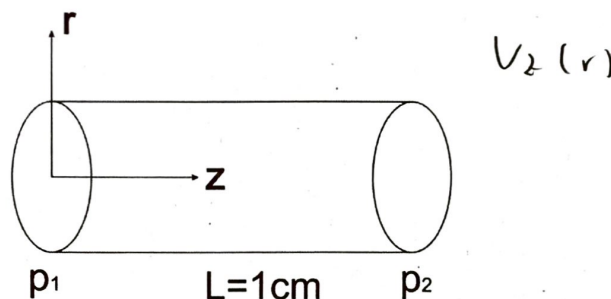
$$= 0.0626 \text{ Pa}$$

$$\tau_{yr} = \frac{p_r - p_{r+\Delta r}}{\Delta r} (y - \frac{h}{2}) = \frac{0.0626 \text{ Pa}}{1 \times 10^{-6} \text{ m}} (y - \frac{200 \times 10^{-6} \text{ m}}{2}) =$$

$$\gamma = \frac{\mu}{\rho r}$$

6. You need to design a piece of pipe flow for a microfluidic system. Consider the flow is at steady state. Neglect the "end effects". The liquid is a Newtonian fluid with constant density and viscosity. $\nu = 2 \times 10^{-4} \text{ m}^2/\text{s}$ is the kinematic viscosity of the fluid. The density of the fluid is $\rho = 0.8 \times 10^3 \text{ kg/m}^3$. The circular pipe is placed horizontal (figure 1). The length of the pipe is $L = 1 \text{ cm}$. Flow is from left to right and driven by a syringe pump. The pressure drop for this piece of pipe is $P_1 - P_2 = 0.25 \text{ N/m}^2$. (hint: $P = P(z)$). The special material used to make this pipe can only afford the maximum shear stress 0.02 N/m^2 before yielding.

- (1) With above yield stress, what is the maximum radius of the tube, R ? (25 points)
(2) What is the maximum velocity? (10 points)



1) ∇^2 - direction

$$\nabla^2 V_z = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V_z}{\partial r} \right) = -\frac{1}{\mu} \frac{dP}{dz}$$

$$\mu \left[\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial V_z}{\partial r} \right) \right] = \frac{dP}{dz} = C_1 \quad \frac{dP}{dz} = C_1 \Rightarrow P = C_1 z + C_2$$

$$\frac{\partial}{\partial r} \left(r \frac{\partial V_z}{\partial r} \right) = \frac{C_1 r}{\mu}$$

$$r \frac{\partial V_z}{\partial r} = \frac{C_1 r^2}{2\mu} + C_2$$

BC: 0 at $r=0$, $\frac{\partial V_z}{\partial r} = \text{finite}$

$$\Rightarrow C_2 = 0$$

$$r \frac{\partial V_z}{\partial r} = \frac{C_1 r^2}{2\mu}$$

$$\frac{\partial V_z}{\partial r} = \frac{C_1 r}{2\mu}$$

$$V_z = \frac{C_1 r^2}{4\mu} + C_3$$

BC: at $r=R$, $V_z = 0$

$$V_z = \frac{-25 r^2}{4\mu} + C_3$$

$$0 = \frac{-25 R^2}{4\mu} + C_3 \Rightarrow C_3 = \frac{25 R^2}{4\mu}$$

1) at $z=0$, $P=P_1$

$$P_1 = C_1 \cdot 0 + C_2$$

2) at $z=L=0.01 \text{ m}$, $P=P_2$

$$P_2 = 0.01 C_1 + C_2$$

$$P_2 = 0.01 C_1 + P_1$$

$$C_1 = 100(P_2 - P_1) = -25 \text{ N/m}^2$$

$$\Rightarrow P = -25 \text{ N/m}^2 z + P_1$$

$$V_z = \frac{25 R^2}{4\mu} \left(1 - \left(\frac{r}{R} \right)^2 \right)$$

$$\tau_{rz} = -\mu \frac{dV_z}{dr} = -\frac{25 R^2}{4} \left(-2 \frac{r}{R^2} \right) = \frac{25}{2} r = 0.02$$

$$R = 0.0016 \text{ m}$$

$$2) V_{\max} = \frac{25 R^2}{4\mu} = 0.0001 \text{ m/s}$$

$$V \cdot \rho$$

$$= ?$$